

Dimension Two Has No Spurious Local Minima for the Operator-Valued Frame Potential

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Source Conjecture

This packet concerns Conjecture 5.1 of Needham–Shonkwiler, *Fusion Frame Homotopy and Tightening Fusion Frames by Gradient Descent*, arXiv:2208.11045. In the authors’ notation, for fixed spectra $\mathbf{r}$, the conjecture states that all local minima of the obvious operator-valued frame potential

$$\text{OFP} : \mathcal{OF}^{\mathbb{K}^d, \mathbf{k}}(\mathbf{r}) \rightarrow \mathbb{R}$$

are global minima.

Nonetheless, running gradient descent from random starting points in $\mathcal{OF}^{\mathbb{K}^d, \mathbf{k}}(\mathbf{r})$ seems to always find minimizers of FFP in practice. Figure 1 shows the value of FFP rapidly decreasing to the global minimum value $\frac{11}{2}$, which is greater than the value $\frac{16}{3}$ that a TFF would have. As observed by Casazza and Fickus [15, p. 17], the minimum is achieved by fusion frames of the form shown on the right of Figure 1, where ℓ_1 and ℓ_2 lie in a plane $\mathcal{Q}$ perpendicular to $\mathcal{S}$ and the lines ℓ_1 , ℓ_2 , and $\mathcal{Q} \cap \mathcal{S}$ correspond to a tight Mercedes–Benz frame for $\mathcal{Q}$.

This suggests that the fusion frame Benedetto–Fickus theorem may hold even for parameters where there are no fusion frames with property $\mathcal{S}$. We also expect our approach to proving Theorem 1.14 will extend to more general spaces $\mathcal{OF}^{\mathbb{K}^d, \mathbf{k}}(\mathbf{r})$ of operator-valued frames with fixed spectra of the P_i , though the details seem more complicated. Hence, we pose the following conjecture:

Conjecture 5.1. *Let $d, N, k_1, \dots, k_N$ be positive integers and fix $\mathbf{r}$. Let $\text{OFP} : \mathcal{OF}^{\mathbb{K}^d, \mathbf{k}}(\mathbf{r}) \rightarrow \mathbb{R}$ be the obvious generalization of FFP to operator-valued frames. Then all local minima of FFP are global minima.*

We expect that, as in the case of classical frames [50], $\mathcal{OF}_{\lambda}^{\mathbb{H}^d, \mathbf{k}}(\mathbf{r})$ is path-connected, where $\mathbb{H}$ is the skew-field of quaternions. However, $\mathcal{OF}_{\lambda}^{\mathbb{R}^d, \mathbf{k}}(\mathbf{r})$ cannot always be connected. For example, translating a result of Kapovich and Millson [35, Theorem 1] to our setting and notation implies that $\mathcal{OF}_{(5,5)}^{\mathbb{R}^2, (1,1,1,1)}(3,3,3,1)$ is not connected. On the other hand, Cahill, Mixon, and Strawn [12] proved that the space $\mathcal{OF}_{(\frac{N}{d}, \dots, \frac{N}{d})}^{\mathbb{R}^d, (1, \dots, 1)}(1, \dots, 1)$ of real unit-norm tight

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Result

We prove the conjecture for $d = 2$, over both $\mathbb{K} = \mathbb{R}$ and $\mathbb{K} = \mathbb{C}$, with arbitrary finite fixed spectral data. This is a partial result: it does not address the higher-dimensional conjecture.

Proof Intuition

In dimension two, a Hermitian operator with fixed eigenvalues is just a scalar matrix plus a fixed-radius traceless part. The traceless part is parametrized by a unit circle in the real case and by a unit sphere in the complex case. Therefore the variable part of OFP is the squared length of a weighted vector sum

$$\left| \sum_i c_i x_i \right|^2, \quad x_i \in S^q, \quad c_i \geq 0.$$

This elementary energy has no spurious local minima. If the weighted sum is zero, the point is globally minimizing. If it is nonzero, first variation forces every x_i to be parallel or antiparallel to the sum. A collinear critical point with two or more vectors pointing along the sum can be lowered by rotating two of them in opposite tangent directions. Hence a nonzero local minimum can only occur when one weight dominates all the others; by the triangle inequality that configuration is already globally minimizing.

Weighted-Sphere Lemma

Lemma 1. *Let $q \geq 1$, let $c_1, \dots, c_N \geq 0$, and define*

$$F : (S^q)^N \rightarrow \mathbb{R}, \quad F(x_1, \dots, x_N) = \left| \sum_{i=1}^N c_i x_i \right|^2.$$

Then every local minimum of F is a global minimum.

Proof. Discarding the indices with $c_i = 0$ changes neither local nor global minimality, so assume $c_i > 0$. Let $x = (x_i)$ be a local minimum and set

$$R = \sum_i c_i x_i.$$

If $R = 0$, then $F(x) = 0$, so x is a global minimum.

Assume $R \neq 0$, and write $u = R/|R|$. The first variation of F in the tangent space $T_{x_i} S^q$ is

$$v_i \mapsto 2c_i \langle R, v_i \rangle.$$

At a local minimum this vanishes for every $v_i \perp x_i$. Hence R is parallel to each x_i , so $x_i = \varepsilon_i u$ with $\varepsilon_i \in \{\pm 1\}$. Put

$$L = \sum_i \varepsilon_i c_i = |R| > 0.$$

We claim that at most one sign ε_i can be $+1$. Suppose instead that $\varepsilon_p = \varepsilon_q = +1$ for two distinct indices p, q . Choose a unit vector $v \perp u$, possible since $q \geq 1$, and choose nonzero real numbers a_p, a_q with $c_p a_p + c_q a_q = 0$. Consider the curve that leaves all x_i fixed except

$$x_p(t) = u \cos(a_p t) + v \sin(a_p t), \quad x_q(t) = u \cos(a_q t) + v \sin(a_q t).$$

The first-order tangent change of R is zero, while the second-order change in the u -direction is

$$-\frac{t^2}{2} (c_p a_p^2 + c_q a_q^2) u + O(t^3).$$

Consequently

$$F(x(t)) = L^2 - L(c_p a_p^2 + c_q a_q^2)t^2 + O(t^3) < L^2$$

for all sufficiently small nonzero t , contradicting local minimality.

Thus exactly one index, say p , has $\varepsilon_p = +1$, and

$$L = c_p - \sum_{i \neq p} c_i > 0.$$

For any $y_i \in S^q$, the triangle inequality gives

$$\left| \sum_i c_i y_i \right| \geq c_p - \sum_{i \neq p} c_i = L.$$

Therefore x attains the global minimum value. This proves the lemma. $\square$

Dimension-Two Theorem

Theorem 1. *Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. In Hilbert dimension $d = 2$, for arbitrary finite fixed spectral data $\mathbf{r}$, every local minimum of the generalized operator-valued frame potential on $\mathcal{OF}^{\mathbb{K}^2, \mathbf{k}}(\mathbf{r})$ is a global minimum.*

Proof. It is enough to work with the positive semidefinite operators

$$P_i = A_i^* A_i$$

with prescribed spectra; the unitary ambiguity in the square roots A_i does not change OFP. Thus

$$\text{OFP}(P_1, \dots, P_N) = \left\| \sum_i P_i \right\|_{\text{HS}}^2 = \text{tr} \left(\left(\sum_i P_i \right)^2 \right).$$

Fix i . Let the two eigenvalues of P_i be $\alpha_i \geq \beta_i \geq 0$, where $\beta_i = 0$ is allowed for rank-one data. Write

$$m_i = \frac{\alpha_i + \beta_i}{2}, \quad c_i = \frac{\alpha_i - \beta_i}{2}.$$

Then

$$P_i = m_i I_2 + c_i X_i,$$

where X_i is self-adjoint, traceless, and has eigenvalues $1, -1$, unless $c_i = 0$, in which case $P_i = m_i I_2$ is constant and can be ignored.

The set of such X_i is naturally a unit sphere: for $\mathbb{K} = \mathbb{C}$, it is the Bloch sphere in the three-dimensional real space of traceless Hermitian 2×2 matrices; for $\mathbb{K} = \mathbb{R}$, it is a unit circle in the two-dimensional real space of traceless symmetric 2×2 matrices. In either case, after choosing an orthonormal basis of the traceless space, write $X_i \leftrightarrow x_i \in S^q$, where $q = 2$ in the complex case and $q = 1$ in the real case.

Now

$$\sum_i P_i = \left(\sum_i m_i \right) I_2 + \sum_i c_i X_i.$$

The scalar and traceless summands are Hilbert–Schmidt orthogonal, so

$$\text{OFP}(P_1, \dots, P_N) = \text{constant} + C \left| \sum_i c_i x_i \right|^2$$

for a fixed positive normalization constant C depending only on the chosen inner-product convention. Multiplication by a positive constant and addition of a constant preserve local and global minimizers. Therefore the local minima of OFP are exactly the local minima of the weighted-sphere energy in Lemma 1. By that lemma, they are global minima. $\square$

Verification Notes

The proof is analytic and self-contained. A short exploratory Hessian search over diagonal critical points in $d = 3$ found no counterexample, but no computational claim is used here and no higher-dimensional conclusion is asserted. The theorem depends only on the elementary two-dimensional orbit parametrization and Lemma 1.

Novelty Check and Limitations

Local run indexes were searched for the arXiv id and close phrases around operator-valued frame potentials, local versus global minima, and the fusion-frame Benedetto–Fickus theorem. Web searches were also run for close variants of those phrases and returned the source paper but no independent proof of this $d = 2$ case. Novelty confidence is moderate pending expert review.

This packet proves only the $d = 2$ case of Conjecture 5.1. The conjecture for $d \geq 3$, including the full operator-valued fixed-spectrum setting, remains open.

References

References

- [1] T. Needham and C. Shonkwiler, *Fusion Frame Homotopy and Tightening Fusion Frames by Gradient Descent*, arXiv:2208.11045, 2022.